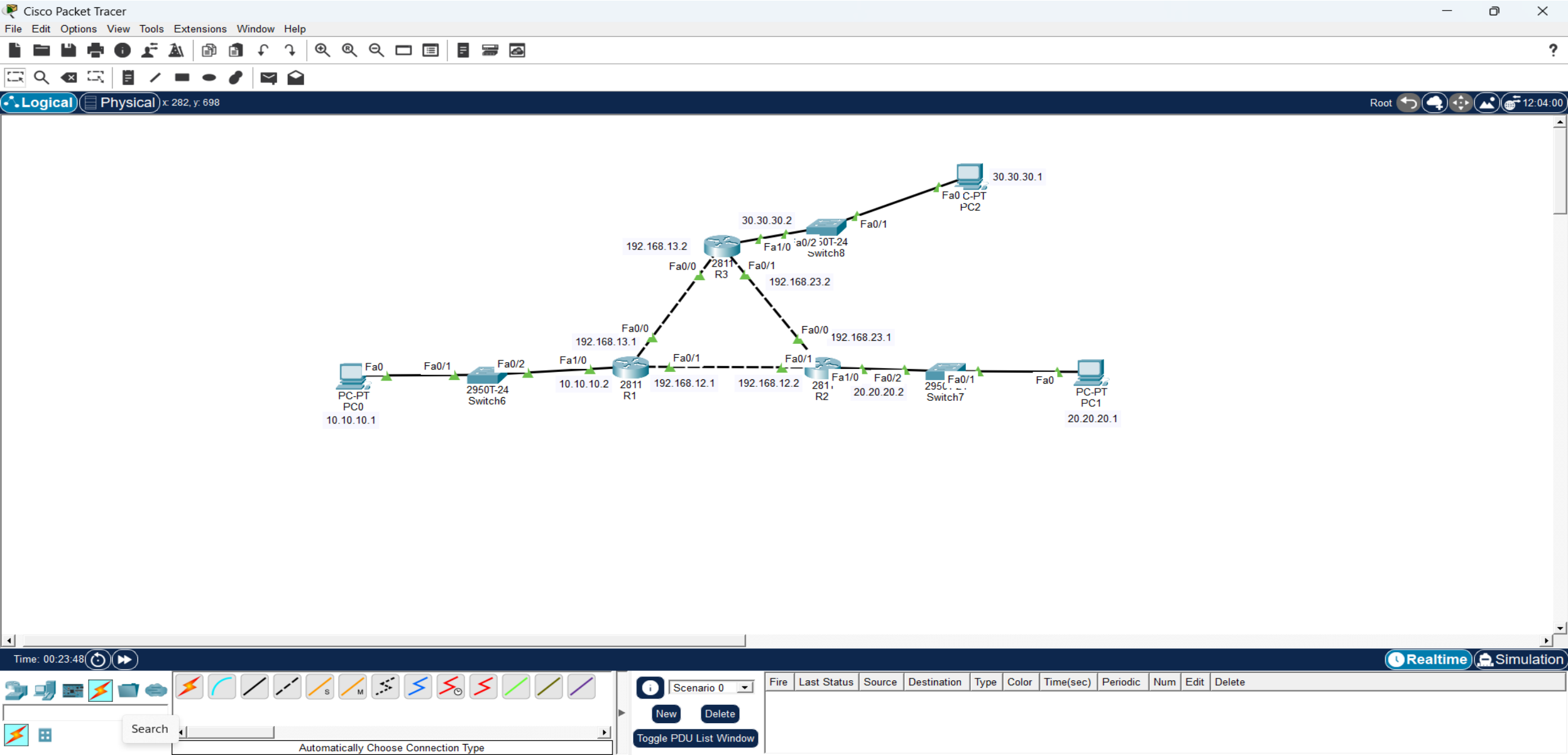
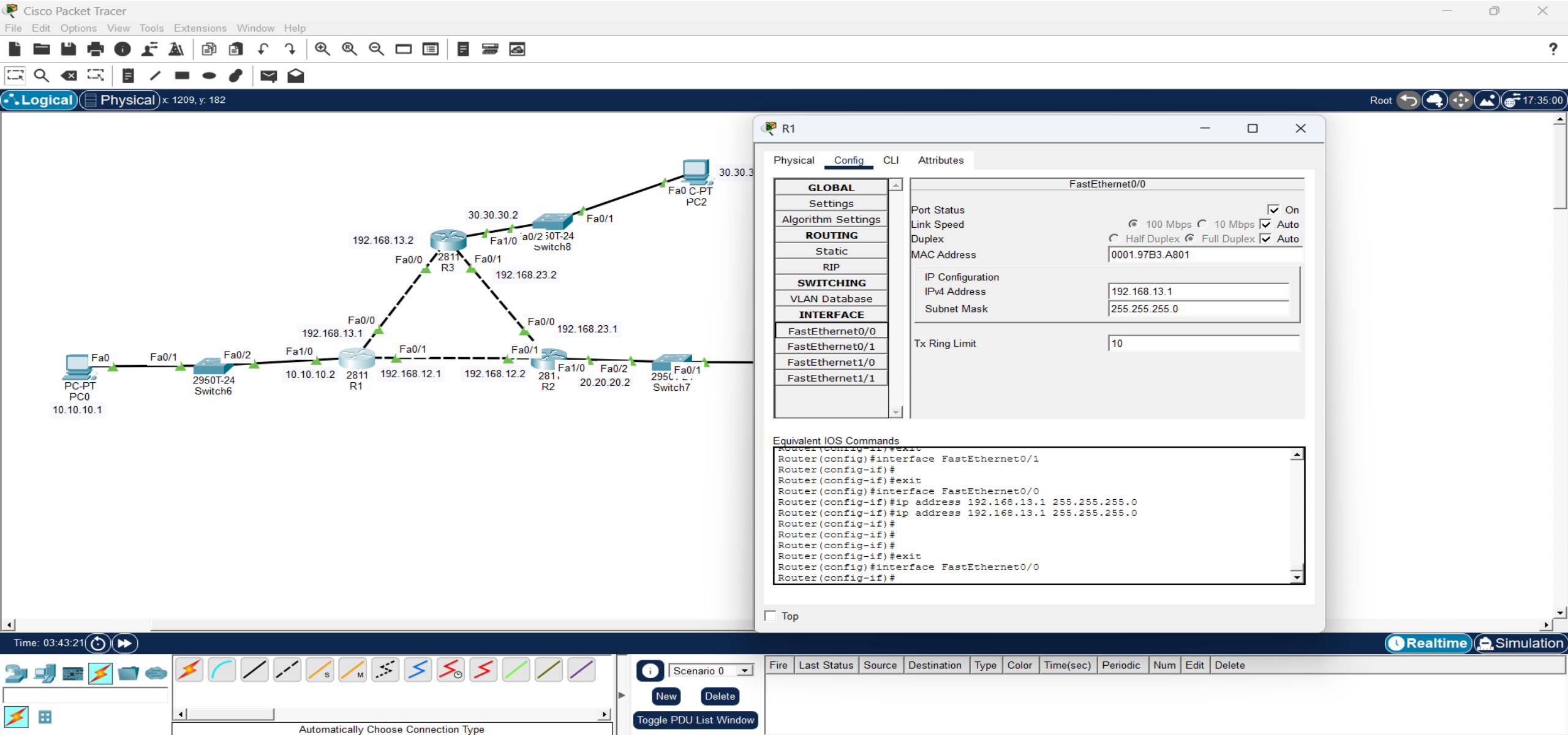
Setting up the Network

- Capt Abhijeet

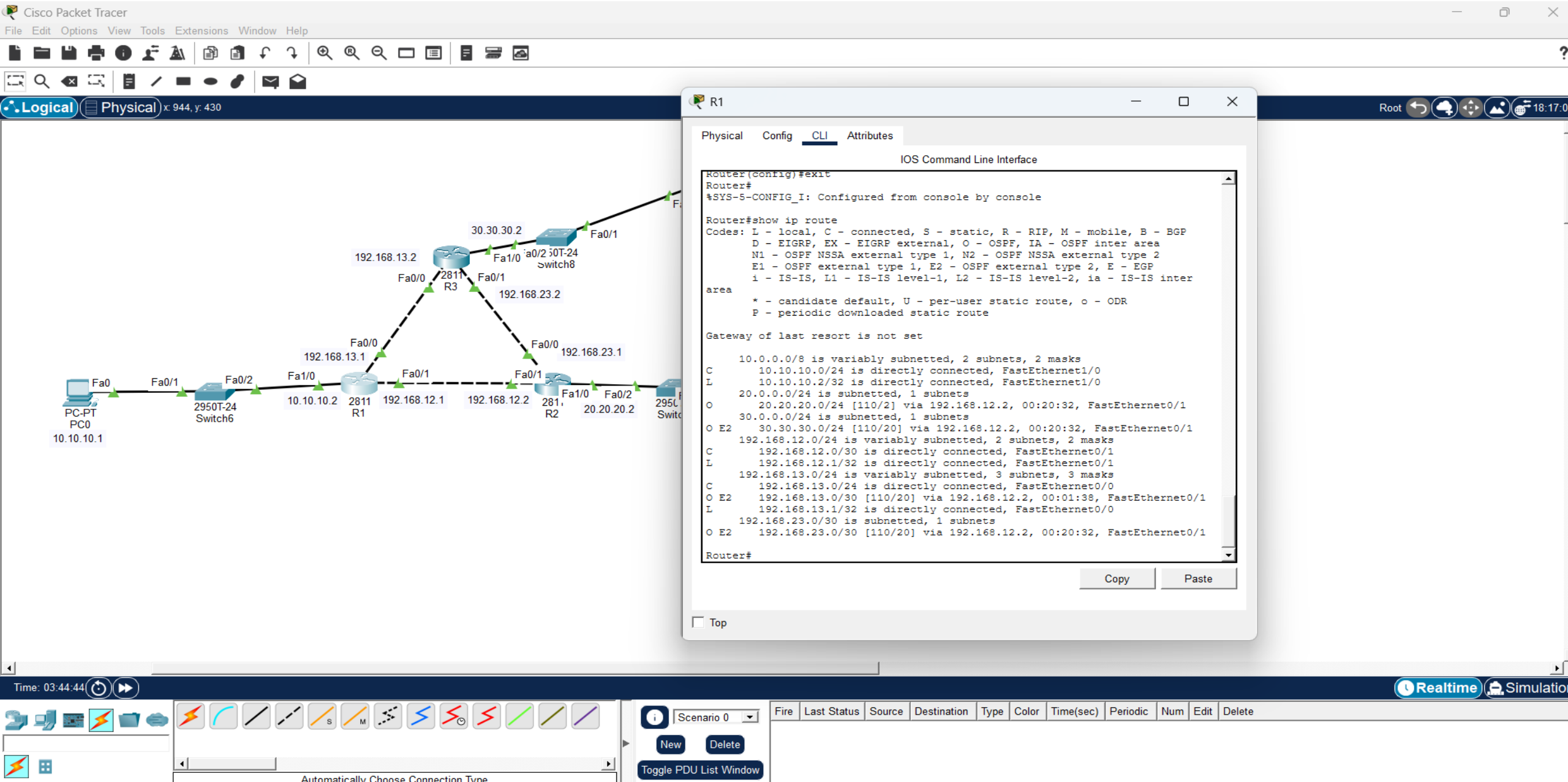


Inputting wrong wildcard mask

- Capt Abhijeet



Connection not seen in ip route - Capt Abhijeet



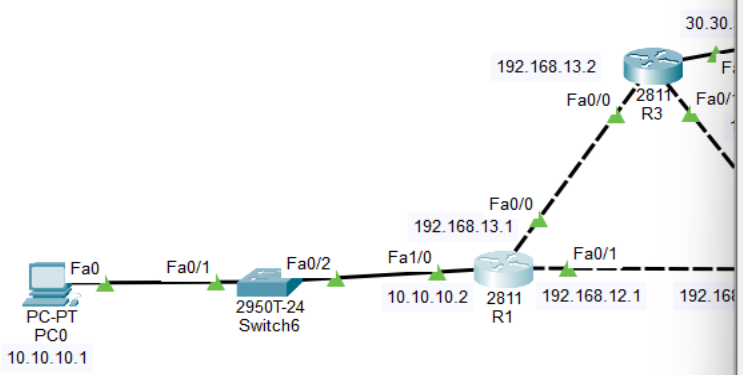
Ping to 30.30.30.1 successful

- Capt Abhijeet

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical | Physical | x: 1832, y: 540 | Root | 18:48:00



R1

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
     20.0.0.0/24 is subnetted, 1 subnets
O       20.20.20.0/24 [110/2] via 192.168.12.2, 00:20:32, FastEthernet0/1
     30.0.0.0/24 is subnetted, 1 subnets
O E2    30.30.30.0/24 [110/20] via 192.168.12.2, 00:20:32, FastEthernet0/1
     192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/30 is directly connected, FastEthernet0/1
L       192.168.12.1/32 is directly connected, FastEthernet0/1
     192.168.13.0/24 is variably subnetted, 3 subnets, 3 masks
C       192.168.13.0/24 is directly connected, FastEthernet0/0
O E2    192.168.13.0/30 [110/20] via 192.168.12.2, 00:01:38, FastEthernet0/1
L       192.168.13.1/32 is directly connected, FastEthernet0/0
     192.168.23.0/30 is subnetted, 1 subnets
O E2    192.168.23.0/30 [110/20] via 192.168.12.2, 00:20:32, FastEthernet0/1

Router#
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.13.1 255.255.255.252
Router(config-if)#
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 30.30.30.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 30.30.30.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#
```

Copy | Paste

Top

Time: 03:45:44 | Realtime | Simulation

Automatically Choose Connection Type

Scenario 0 | New | Delete | Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete

Passive interface

- Capt Abhijeet

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical | Physical x: 623, y: 27 | Root | 21:51:00

R1

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
00:48:34: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.23.1 on FastEthernet0/1 from
FULL to DOWN, Neighbor Down: Interface down or detached
no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed
state to up

Router(config-if)#
Router(config-if)#conf t
%Invalid hex value
Router(config)#router ospf 1
Router(config-router)#passive
00:49:33: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.23.1 on FastEthernet0/1 from
LOADING to FULL, Loadin
Router(config-router)#
Router(config-router)#passive-interface FastEthernet0/1
Router(config-router)#
00:49:50: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.23.1 on FastEthernet0/1 from
FULL to DOWN, Neighbor Down: Interface down or detached

Router(config-router)#show ip ospf neighbor
                     ^
% Invalid input detected at '^' marker.

Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip ospf neighbor

Router#
```

Copy | Paste

Top

30.30.30.1 Fa0 C-PT PC2

PC-P PC0 10.10.1

Fa0/1 witch7

Time: 03:51:44 | Realtime | Simulation

Scenario | Fire | Last Sta | New | Delete | PDU List W

Automatically Choose Connection Type

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical | Physical x: 1172, y: 459 | Root | 22:23:30

R1

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
00:51:18: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.23.1 on FastEthernet0/1 from
LOADING to FULL, Loading Done

 Routing Process "ospf 1" with ID 192.168.13.1
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 3. Checksum Sum 0x01e984
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0)
        Number of interfaces in this area is 3
        Area has no authentication
        SPF algorithm executed 15 times
        Area ranges are
        Number of LSA 3. Checksum Sum 0x01e262
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0

Router#
Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.23.1      1   FULL/DR         00:00:39    192.168.12.2
FastEthernet0/1
Router#
```

Copy | Paste

Top

30.30.30.1 Fa0 C-PT PC2

PC-P PC0 10.10.1

Fa0/1 witch7

Time: 03:52:47 | Realtime | Simulation

Scenario | Fire | Last Sta | New | Delete | PDU List W

Automatically Choose Connection Type

Wrong Autonomous System

- Capt Abhijeet

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

R3

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#conf t
Enter configuration commands, one per line.   End with CNTL/Z.
Router(config)#no router eigrp 200
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H    Address          Interface        Hold Uptime     SRTT    RTO    Q    Seq
                                       (sec)           (ms)           Cnt  Num
0    192.168.23.1     Fa0/1            11   00:34:42   40      1000   0    63

Router#conf t
Enter configuration commands, one per line.   End with CNTL/Z.
Router(config)#router eigrp 20
Router(config-router)#exit
Router(config)#show ip eigrp neighbors
                ^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H    Address          Interface        Hold Uptime     SRTT    RTO    Q    Seq
                                       (sec)           (ms)           Cnt  Num
0    192.168.23.1     Fa0/1            12   00:35:27   40      1000   0    63

IP-EIGRP neighbors for process 20

Router#
```

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Top

Time: 03:56:44

Realtime Simulation

Automatically Choose Connection Type

Missing Network Statement

- Capt Abhijeet

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical | Physical x: 743, y: 486 | Root | 02:26:00

R2

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
state to down

00:53:26: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.13.1 on FastEthernet0/1 from
FULL to DOWN, Neighbor Down: Interface down or detached

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed
state to up

00:54:25: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.13.1 on FastEthernet0/1 from
LOADING to FULL, Loading Done

00:55:20: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.13.1 on FastEthernet0/1 from
FULL to DOWN, Neighbor Down: Dead timer expired

00:55:20: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.13.1 on FastEthernet0/1 from
FULL to DOWN, Neighbor Down: Interface down or detached

00:56:10: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.13.1 on FastEthernet0/1 from
LOADING to FULL, Loading Done


Router>
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#no network 20.20.20.0 0.0.0.255 area 0
Router(config-router)#show ip route
                          ^
% Invalid input detected at '^' marker.

Router(config-router)#exit
Router(config)#exit
Router#
```

Copy | Paste

Top

Time: 04:00:43 | Simulation

Scenario | New | Delete | PDU List W | Fire | Last Sta

Automatically Choose Connection Type

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical | 02:45:00

R1

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Router#
Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.23.1      1   FULL/DR         00:00:39    192.168.12.2
FastEthernet0/1
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.10.0/24 is directly connected, FastEthernet1/0
L       10.10.10.2/32 is directly connected, FastEthernet1/0
     30.0.0.0/24 is subnetted, 1 subnets
O E2    30.30.30.0/24 [110/20] via 192.168.12.2, 00:08:36, FastEthernet0/1
     192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/30 is directly connected, FastEthernet0/1
L       192.168.12.1/32 is directly connected, FastEthernet0/1
     192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.13.0/30 is directly connected, FastEthernet0/0
L       192.168.13.1/32 is directly connected, FastEthernet0/0
     192.168.23.0/30 is subnetted, 1 subnets
O E2    192.168.23.0/30 [110/20] via 192.168.12.2, 00:08:36, FastEthernet0/1

Router#
```

Copy | Paste

Top

Time: 04:01:20 | Realtime | Simulation

Scenario | New | Delete | PDU List W | Fire | Last Sta

Automatically Choose Connection Type

Missing Network Statement

- Capt Abhijeet

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.10.0/24 is directly connected, FastEthernet1/0
L       10.10.10.2/32 is directly connected, FastEthernet1/0
     30.0.0.0/24 is subnetted, 1 subnets
O E2    30.30.30.0/24 [110/20] via 192.168.12.2, 00:08:36, FastEthernet0/1
     192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/30 is directly connected, FastEthernet0/1
L       192.168.12.1/32 is directly connected, FastEthernet0/1
     192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.13.0/30 is directly connected, FastEthernet0/0
L       192.168.13.1/32 is directly connected, FastEthernet0/0
     192.168.23.0/30 is subnetted, 1 subnets
O E2    192.168.23.0/30 [110/20] via 192.168.12.2, 00:08:36, FastEthernet0/1

Router#show ip protocol

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.13.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.255 area 0
    192.168.12.0 0.0.0.3 area 0
    192.168.13.0 0.0.0.3 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    192.168.13.1         110      00:09:08
    192.168.23.1         110      00:01:25
  Distance: (default is 110)

Router#
```

Copy Paste

Top

Time: 04:01:54

Realtime Simulation

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
     192.168.13.0 0.0.0.3 area 0
     20.20.20.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    192.168.13.1         110      00:10:43
    192.168.23.1         110      00:03:00
  Distance: (default is 110)

Router>conf t
          ^
% Invalid input detected at '^' marker.

Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#no network 20.20.20.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
pi
Protocol [ip]:
Target IP address: ping 20.20.20.1
% Bad IP address

Router#ping 20.20.20.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.20.20.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#
```

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Top

Time: 04:04:58

Realtime Simulation

DELIVERABLES

- Capt Abhijeet

- Wrong Wildcard Mask
  - “Incorrect wildcard mask caused OSPF to not match the correct interface, preventing neighbor adjacency.”
  - Commands used : show ip protocol
- Passive Interface
  - “Passive interface suppressed OSPF hello packets, preventing neighbor formation.”
  - Command used: show ip ospf neighbor
- Wrong Autonomous System
  - “EIGRP adjacency failed due to Autonomous System mismatch.”
  - Command used: show ip eigrp neighbors, show ip protocol
- Missing Network Statement
  - “Network was not advertised into OSPF due to missing network statement.”
  - Command used: show ip route, show ip protocol